

SMARTPHONE SoCs

SMARTPHONE SOCS ARE ESSENTIALLY TINY PRE-CONFIGURED COMPUTERS EMBEDDED WITHIN YOUR DEVICES.

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So far we have looked at the processors and graphics chips in computers and laptops, but in smartphones, these choices are not made by the end user, and for the most part, not even by the OEM. The System on a Chip (SoC) companies call the shots, and decide the design and configuration. What they vend is essentially a tiny pre made computer with the CPU and GPU on a single chip. Within a phone, the entire SoC is referred to as a processor, the complexes of CPUs on an SoC are also called processors, each individual CPU is a processor, the individual cores within the CPUs are also processors. This is like a processorception, so let's start at the top.

THE SOCS

All the mobile manufacturers use various generations of a handful of SoCs, which are referred to as families. These few families of SoCs together rule the roost. The most widely used one by far is the Snapdragon family by Qualcomm. Exynos by Samsung, Kirin by HiSilicon, Ax by Apple, and Helio by MediaTek are the others. Snapdragon and MediaTek are used by a number of smartphone brands. Kirin and Ax are used exclusively in house by Huawei and Apple. Exynos SoCs are

mostly used in house by Samsung, but a handful of Meizu phones use them as well. The latest generation of almost all these SoCs are manufactured by the 7nm node of a single company, the Taiwan Semiconductor Manufacturing Company (TSMC). Samsung is the only outlier, and fabricates its Exynos SoCs with an 8nm process. All the SoCs, without exception, are manufactured using the CMOS process. The

Kirin 980, the Exynos 9820, the Apple A12, the Helio P90 and the Snapdragon 855 are the particular SoCs that are the latest generation in their respective families. Most of the newer generations of SoCs have a few basic considerations. They should support the next generation of communications technology, which is 5G, they should consume as little electricity as possible to extend the battery life, be power-



The Qualcomm 855 block diagram

Credit: Qualcomm